Cognitive Diseases Pre-Screening Framework

Version	1.0	**Date**	June 2025
Target Users	Adults 18+ (50+ primary focus)	**Clinical Basis**	PANDA, Hachinski Scale, DLB Consortium, FBI, UHDRS, BICAMS
Diseases Covered	6 cognitive conditions	**Document Purpose**	Pre-screening questionnaire & scoring guide

Clinical Disclaimer: This document is a pre-screening triage tool only. It is not a validated diagnostic instrument and cannot diagnose, confirm, or rule out any neurological or cognitive condition. Questions and scoring weights are drawn from publicly available, validated clinical instruments. A high score does not mean a person has the disease. A low score does not rule it out. Only a qualified clinician can perform a full assessment and provide a diagnosis. Independent clinical validation and ethics review are strongly recommended before any public deployment.

Contents

What is it?	Parkinson's disease is a brain condition that mainly affects movement. Over time, many people with Parkinson's also develop thinking and memory problems. This is called Parkinson's disease dementia (PDD). About 30% of people with Parkinson's will develop dementia, and up to 80% will have some level of cognitive change during the course of the disease.
What it affects	Attention, visuospatial skills, executive function, processing speed, memory. Motor symptoms (tremor, stiffness, slow movement) come first; cognitive symptoms usually follow later in the disease.
Screening tools	PANDA (Parkinson Neuropsychometric Dementia Assessment), MoCA-PD (Montreal Cognitive Assessment weighted for Parkinson's), PD-CRS (Parkinson's Disease Cognitive Rating Scale)

Questions

Answer Options and Scores	Never = 0	Occasionally (1-2x/month) = 1	Frequently (weekly) = 2	Getting noticeably worse = 3

#	Question	Never (0)	Occasionally (1)	Frequently (2)	Getting worse (3)
PD1	I notice my thinking feels slower than it used to be, even for simple tasks.	[ ]	[ ]	[ ]	[ ]
PD2	I have trouble concentrating on a task or a conversation for more than a few minutes.	[ ]	[ ]	[ ]	[ ]
PD3	I misjudge distances, bump into things, or find parking or steps difficult.	[ ]	[ ]	[ ]	[ ]
PD4	I find it hard to switch between two tasks or follow two things at once.	[ ]	[ ]	[ ]	[ ]
PD5	I know what I want to say but the word does not come out, even for everyday words.	[ ]	[ ]	[ ]	[ ]
PD6	I physically act out my dreams during sleep, such as kicking, punching, or shouting.	[ ]	[ ]	[ ]	[ ]
PD7	I feel emotionally flat or have lost interest in most things I used to enjoy.	[ ]	[ ]	[ ]	[ ]
PD8	I see things that other people cannot see, such as people, animals, or objects that are not there. **[RED FLAG]**	[ ]	[ ]	[ ]	[ ]
PD9	Someone close to me has said my memory or thinking seems worse recently.	[ ]	[ ]	[ ]	[ ]
PD10	I have trouble remembering what I was doing partway through a task.	[ ]	[ ]	[ ]	[ ]

Red Flag Note: Question PD8 (visual hallucinations) is a strong clinical warning sign. A score of 2 or 3 should automatically raise the result to at least Moderate Concern and prompt an urgent GP referral, as this symptom can also indicate Lewy Body Dementia.

Domain Weights

Domain	Weight	Max Raw Score	Questions
Attention and Processing Speed	30%	9	PD1, PD2, PD10
Visuospatial and Motor-Cognitive	25%	9	PD3, PD4, PD6
Language and Word Finding	20%	6	PD5
Mood and Behaviour	15%	6	PD7, PD8
Third-Party Observation	10%	3	PD9 (1.5x weight)

Scoring Guide			
Score Range	**Label**	**Colour**	**Recommended Action**
0 to 20	Low Concern	Green	No significant signs. Recheck in 12 months.
21 to 45	Mild Concern	Yellow	Some signs present. Mention at next GP visit.
46 to 65	Moderate Concern	Orange	Multiple signs. Book a GP appointment this month.
66 to 100	High Concern	Red	Strong signs. See a doctor soon.

References

[1] Kalbe E, et al. Screening for cognitive deficits in Parkinson's disease with the PANDA instrument. Parkinsonism Relat Disord. 2008;14(2):93-101. DOI: 10.1016/j.parkreldis.2007.06.008 PubMed: https://pubmed.ncbi.nlm.nih.gov/17707678/

[2] Nasreddine ZS, et al. The Montreal Cognitive Assessment (MoCA). J Am Geriatr Soc. 2005;53(4):695-699. https://mocacognition.com/

[3] Galvin JE, et al. Cognitive symptoms in Parkinson disease cross-sectional cohort. PMC: https://www.ncbi.nlm.nih.gov/pmc/articles/PMC11221914/

[4] Gill DJ, et al. Screening for cognitive impairment in PD: improving MoCA through subtest weighting. PLOS One 2016. PMC: https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4954721/

What is it?	Vascular dementia is the second most common type of dementia. It happens when the blood supply to parts of the brain is reduced or blocked, for example due to a stroke or a series of small strokes (TIAs). The brain cells in the affected areas are damaged or die, which causes thinking problems. Unlike Alzheimer's disease, memory is not always the first thing affected. Instead, problems with planning, organising, and thinking speed are often noticed first.
What it affects	Executive function, processing speed, attention, and planning. Memory may be less affected early on. Physical stroke risk factors like high blood pressure, atrial fibrillation, and diabetes are important context.
Screening tools	Hachinski Ischaemic Scale (HIS, 1975), NINDS-AIREN criteria, MoCA (Montreal Cognitive Assessment), Q*mci screen

Questions

Answer Options and Scores	Never = 0	Occasionally (1-2x/month) = 1	Frequently (weekly) = 2	Getting noticeably worse = 3

#	Question	Never (0)	Occasionally (1)	Frequently (2)	Getting worse (3)
VD1	My thinking feels clearly slower than it used to be.	[ ]	[ ]	[ ]	[ ]
VD2	I struggle to plan or organise tasks that have more than a couple of steps.	[ ]	[ ]	[ ]	[ ]
VD3	I lose track of what I was doing or saying in the middle of a task.	[ ]	[ ]	[ ]	[ ]
VD4	I cry or laugh suddenly for no clear reason, even when I do not feel sad or happy.	[ ]	[ ]	[ ]	[ ]
VD5	I have been told I have high blood pressure, diabetes, or atrial fibrillation (irregular heartbeat).	[ ]	[ ]	[ ]	[ ]
VD6	I have had a stroke or a TIA (mini-stroke) in the past. **[RED FLAG]**	[ ]	[ ]	[ ]	[ ]
VD7	My thinking problems came on suddenly, or got noticeably worse in a short space of time. **[RED FLAG]**	[ ]	[ ]	[ ]	[ ]
VD8	I walk with a shuffle, or I have had unexplained falls recently.	[ ]	[ ]	[ ]	[ ]
VD9	I find it hard to follow a conversation or keep up with what is happening around me.	[ ]	[ ]	[ ]	[ ]
VD10	Someone who knows me well has said my personality or thinking has changed.	[ ]	[ ]	[ ]	[ ]

Red Flag Note: Questions VD6 (history of stroke or TIA) and VD7 (sudden onset) are strong clinical red flags. A score of 2 or 3 on either one automatically raises the result to at least Moderate Concern. Sudden or stepwise changes in thinking are a hallmark of vascular dementia and should prompt an urgent medical review.

Domain Weights

Domain	Weight	Max Raw Score	Questions
Executive Function and Planning	32%	9	VD2, VD3, VD9
Processing Speed	25%	6	VD1, VD7
Vascular Risk Factors	20%	6	VD5, VD6 (1.5x weight)
Mood and Behaviour	13%	6	VD4, VD10
Motor and Gait	10%	3	VD8

Scoring Guide			
Score Range	**Label**	**Colour**	**Recommended Action**
0 to 20	Low Concern	Green	No significant signs. Recheck in 12 months.
21 to 45	Mild Concern	Yellow	Some signs present. Mention at next GP visit.
46 to 65	Moderate Concern	Orange	Multiple signs. Book a GP appointment this month.
66 to 100	High Concern	Red	Strong signs. See a doctor soon.

References

[1] Hachinski VC, et al. Cerebral blood flow in dementia. Arch Neurol. 1975;32(9):632-637. DOI: 10.1001/archneur.1975.00490510088009 PubMed: https://pubmed.ncbi.nlm.nih.gov/1138815/

[2] O'Caoimh R, et al. Comparing diagnostic accuracy of screening tools across dementia subtypes. PMC: https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6787633/

[3] Roman GC, et al. Vascular dementia: diagnostic criteria for research studies (NINDS-AIREN). Neurology. 1993;43(2):250-260. DOI: 10.1212/WNL.43.2.250

[4] Iadecola C, et al. Vascular cognitive impairment and dementia. Nat Rev Dis Primers. 2019;5(1):21. DOI: 10.1038/s41572-019-0064-2

y Bodies

DLB

What is it?	Dementia with Lewy Bodies (DLB) is the second most common form of degenerative dementia after Alzheimer's disease. It is caused by small protein deposits called Lewy bodies forming inside brain cells. DLB has four core features: thinking ability that goes up and down (fluctuating cognition), seeing things that are not there (visual hallucinations), REM sleep behaviour disorder (acting out dreams), and movement problems similar to Parkinson's. A diagnosis of probable DLB requires two core features, or one core feature plus a supporting biomarker.
What it affects	Fluctuating attention and alertness, visuospatial skills, and executive function. Unlike Alzheimer's, memory is often less affected in the early stages. Visual hallucinations and REM sleep disorder are particularly distinctive.
Screening tools	DLB Consortium Fourth Consensus Criteria (McKeith et al., 2017, Neurology), MoCA (with visuospatial domain weighted more heavily)

Questions

Answer Options and Scores	Never = 0	Occasionally (1-2x/month) = 1	Frequently (weekly) = 2	Getting noticeably worse = 3

#	Question	Never (0)	Occasionally (1)	Frequently (2)	Getting worse (3)
DL1	My alertness and thinking ability varies a lot hour to hour or day to day. Sometimes I seem fine; other times I am confused.	[ ]	[ ]	[ ]	[ ]
DL2	I see detailed visions of people, animals, or objects that other people cannot see. **[RED FLAG]**	[ ]	[ ]	[ ]	[ ]
DL3	I physically act out my dreams during sleep, such as punching, kicking, or shouting. **[RED FLAG]**	[ ]	[ ]	[ ]	[ ]
DL4	I have difficulty recognising familiar faces or understanding what I am looking at.	[ ]	[ ]	[ ]	[ ]
DL5	I have a tremor at rest, or my movements have become slow and stiff.	[ ]	[ ]	[ ]	[ ]
DL6	I have had unexplained falls or fainting spells.	[ ]	[ ]	[ ]	[ ]
DL7	I struggle to concentrate or pay attention, and this changes noticeably from day to day.	[ ]	[ ]	[ ]	[ ]
DL8	I feel more anxious or low in mood than before, in a way that is new for me.	[ ]	[ ]	[ ]	[ ]
DL9	I have trouble judging distances or navigating spaces I know well.	[ ]	[ ]	[ ]	[ ]
DL10	Someone who knows me well has noticed my alertness or behaviour varies a lot day to day.	[ ]	[ ]	[ ]	[ ]

Red Flag Note: Questions DL2 (visual hallucinations) and DL3 (REM sleep behaviour disorder) are core diagnostic features of DLB according to McKeith et al. (2017). A score of 2 or 3 on either one automatically raises the result to at least Moderate Concern. If both are scored 2 or more, escalate to High Concern regardless of the total score. This combination is a strong indicator that warrants urgent specialist review.

Domain Weights

Domain	Weight	Max Raw Score	Questions
Fluctuating Cognition and Attention	30%	9	DL1, DL7, DL10
Core DLB Features	30%	6	DL2 (1.5x), DL3 (1.5x)
Visuospatial and Perception	20%	6	DL4, DL9
Parkinsonism and Falls	10%	6	DL5, DL6
Mood and Behaviour	10%	6	DL8

Scoring Guide

Score Range	Label	Colour	Recommended Action
0 to 20	Low Concern	Green	No significant signs. Recheck in 12 months.
21 to 45	Mild Concern	Yellow	Some signs present. Mention at next GP visit.
46 to 65	Moderate Concern	Orange	Multiple signs. Book a GP appointment this month.
66 to 100	High Concern	Red	Strong signs. See a doctor soon.

References

[1] McKeith IG, et al. Diagnosis and management of dementia with Lewy bodies: Fourth consensus report of the DLB Consortium. Neurology. 2017;89(1):88-100. PMC: https://www.ncbi.nlm.nih.gov/pmc/articles/PMC5496518/

[2] Lewy Body Dementia Association. Updated diagnostic criteria for DLB. https://lbda.org/new-diagnostic-criteria-published-for-dlb

[3] Boot BP. Comprehensive treatment of dementia with Lewy bodies. Alzheimers Res Ther. 2015;7(1):45. PMC: https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4437654/

[4] Nasreddine ZS, et al. The Montreal Cognitive Assessment. J Am Geriatr Soc. 2005;53(4):695-699.

What is it?	Frontotemporal dementia (FTD) is a group of conditions caused by damage to the frontal and temporal lobes of the brain. Unlike Alzheimer's disease, memory is usually not the first thing affected. Instead, the main early signs involve changes in personality, behaviour, and language. FTD is the most common form of dementia in people under 65. There are two main types: the behavioural variant (bvFTD), which mainly causes personality and behaviour changes, and the language variant (Primary Progressive Aphasia, PPA), which mainly affects speech and word understanding.
What it affects	Personality and behaviour (disinhibition, apathy, loss of empathy), language and speech, executive function. Memory and visuospatial skills are less affected in early stages. Most common dementia in the under-65 age group.
Screening tools	Frontal Behavioural Inventory (FBI, Kertesz et al., 1997, Can J Neurol Sci 24:29-36), Cambridge Behavioural Inventory Revised (CBI-R)

Questions

Answer Options and Scores	Never = 0	Occasionally (1-2x/month) = 1	Frequently (weekly) = 2	Getting noticeably worse = 3

#	Question	Never (0)	Occasionally (1)	Frequently (2)	Getting worse (3)
FT1	I behave in ways that are socially not appropriate, for example making rude comments, ignoring personal space, or acting on impulse in public.	[ ]	[ ]	[ ]	[ ]
FT2	I have lost all motivation and sit for hours doing nothing, even when asked to do something.	[ ]	[ ]	[ ]	[ ]
FT3	I follow the same rigid routines every day and get very upset if they are changed or interrupted.	[ ]	[ ]	[ ]	[ ]
FT4	I have developed a sudden craving for sweet or starchy foods, or I eat much more than before.	[ ]	[ ]	[ ]	[ ]
FT5	Others have said I seem cold, unfeeling, or uncaring in a way I never was before. **[RED FLAG]**	[ ]	[ ]	[ ]	[ ]
FT6	I am losing the meaning of everyday words. I no longer understand what some common objects are called.	[ ]	[ ]	[ ]	[ ]
FT7	My personality feels very different from how I used to be, in a way I cannot explain. **[RED FLAG]**	[ ]	[ ]	[ ]	[ ]
FT8	I repeat the same words, phrases, or actions again and again without realising it.	[ ]	[ ]	[ ]	[ ]
FT9	I have difficulty planning or organising even simple tasks, like making a meal or getting dressed.	[ ]	[ ]	[ ]	[ ]
FT10	Friends or family have said my behaviour has changed in a way that is very out of character.	[ ]	[ ]	[ ]	[ ]

Red Flag Note: Questions FT5 (loss of empathy) and FT7 (personality change) are key diagnostic indicators from the FBI (Kertesz et al., 1997). For anyone under the age of 65, a score of 2 or 3 on either question should automatically raise the result to at least Moderate Concern. Significant personality change in a younger adult is the most important early warning sign of FTD and should not be attributed to stress or depression without medical review.

Domain Weights

Domain	Weight	Max Raw Score	Questions
Behaviour and Disinhibition	30%	12	FT1, FT3, FT8, FT10
Apathy and Motivation Loss	25%	6	FT2, FT4
Loss of Empathy and Personality	20%	6	FT5 (1.5x), FT7 (1.5x)
Language and Semantics	15%	3	FT6
Executive Function	10%	3	FT9

Scoring Guide

Score Range	Label	Colour	Recommended Action
0 to 20	Low Concern	Green	No significant signs. Recheck in 12 months.
21 to 45	Mild Concern	Yellow	Some signs present. Mention at next GP visit.
46 to 65	Moderate Concern	Orange	Multiple signs. Book a GP appointment this month.
66 to 100	High Concern	Red	Strong signs. See a doctor soon.

References

[1] Kertesz A, Davidson W, Fox H. Frontal behavioral inventory: diagnostic criteria for frontal lobe dementia. Can J Neurol Sci. 1997;24(1):29-36. PubMed: https://pubmed.ncbi.nlm.nih.gov/9043744/

[2] Kertesz A, et al. The Frontal Behavioral Inventory in the differential diagnosis of frontotemporal dementia. J Int Neuropsychol Soc. 2000;6(4):460-468. PubMed: https://pubmed.ncbi.nlm.nih.gov/10902415/

[3] Rascovsky K, et al. Sensitivity of revised diagnostic criteria for bvFTD. Brain. 2011;134(9):2456-2477. DOI: 10.1093/brain/awr179

[4] Pijnenburg YA, et al. Behavioural disturbances differentiate FTLD subtypes and AD. Int J Geriatr Psychiatry. 2013. DOI: 10.1002/gps.3907

What is it?	Huntington's disease (HD) is a genetic condition in which certain nerve cells in the brain break down over time. It is caused by a change (mutation) in the HTT gene. Each child of a parent with HD has a 50% chance of inheriting the gene. HD affects movement, thinking, and mood. Involuntary jerky movements called chorea are the most well-known physical sign. Cognitive problems often appear before or alongside motor symptoms and include slowed thinking, difficulty planning, and trouble with attention and working memory.
What it affects	Processing speed (most affected), executive function, attention, working memory. Emotional dysregulation and depression are also very common. Family history of HD is a critical risk factor.
Screening tools	UHDRS Cognitive Battery (Unified Huntington's Disease Rating Scale), PD-CRS (Parkinson's Disease Cognitive Rating Scale, validated for HD by Borroni et al. 2023, PMC10576674), Symbol Digit Modalities Test (SDMT)

Questions

Answer Options and Scores Never = 0 Occasionally (1-2x/month) = 1 Frequently (weekly) = 2 Getting noticeably worse = 3

#	Question	Never (0)	Occasionally (1)	Frequently (2)	Getting worse (3)
HD1	My thinking and responses feel noticeably slower. Others have commented I take longer to answer.	[ ]	[ ]	[ ]	[ ]
HD2	I find it hard to organise my day or complete tasks in the right order.	[ ]	[ ]	[ ]	[ ]
HD3	I have difficulty focusing on something for more than a few minutes without losing track.	[ ]	[ ]	[ ]	[ ]
HD4	I have sudden outbursts of anger, sadness, or irritability that seem out of proportion.	[ ]	[ ]	[ ]	[ ]
HD5	I have noticed involuntary jerky or twitching movements in my hands, arms, or face. **[RED FLAG]**	[ ]	[ ]	[ ]	[ ]
HD6	A parent or sibling has been diagnosed with Huntington's disease. **[RED FLAG]**	[ ]	[ ]	[ ]	[ ]
HD7	I have trouble holding information in my head while I am in the middle of doing something.	[ ]	[ ]	[ ]	[ ]
HD8	I feel persistently sad, low, or empty in a way that is unusual for me.	[ ]	[ ]	[ ]	[ ]
HD9	I make impulsive decisions or act without thinking, in a way I did not before.	[ ]	[ ]	[ ]	[ ]
HD10	My handwriting, speech, or ability to do fine movements has gotten worse.	[ ]	[ ]	[ ]	[ ]

Red Flag Note: Question HD6 (family history of Huntington's disease) is the most important risk indicator. Any score of 1 or more on HD6, combined with a score of 2 or more on HD5 (involuntary movements), should automatically raise the result to High Concern. This combination strongly suggests the person should speak to a doctor about genetic testing and counselling. Genetic testing for HD can confirm whether someone carries the mutation.

Domain Weights

Domain	Weight	Max Raw Score	Questions
Processing Speed and Attention	30%	9	HD1, HD3, HD7
Executive Function	25%	6	HD2, HD9
Motor Symptoms	20%	6	HD5, HD10
Mood and Emotional Control	15%	6	HD4, HD8
Family History (Risk Factor)	10%	3	HD6 (2x weight)

Scoring Guide			
Score Range	**Label**	**Colour**	**Recommended Action**
0 to 20	Low Concern	Green	No significant signs. Recheck in 12 months.
21 to 45	Mild Concern	Yellow	Some signs present. Mention at next GP visit.
46 to 65	Moderate Concern	Orange	Multiple signs. Book a GP appointment this month.
66 to 100	High Concern	Red	Strong signs. See a doctor soon.

References

[1] Borroni B, et al. Measuring cognitive impairment and monitoring cognitive decline in Huntington's disease: a comparison of assessment instruments (PD-CRS validation for HD). PMC: https://www.ncbi.nlm.nih.gov/pmc/articles/PMC10576674/

[2] Tabrizi SJ, et al. Biological and clinical manifestations of Huntington's disease. Lancet Neurol. 2009;8(1):83-94. DOI: 10.1016/S1474-4422(08)70278-X

[3] Unified Huntington's Disease Rating Scale (UHDRS). Huntington Study Group. Mov Disord. 1996;11(2):136-142. DOI: 10.1002/mds.870110204

[4] Cognitive Assessment Tools for Huntington's Disease. ClinicalTrials.gov. https://clinicaltrials.gov/study/NCT06546488

What is it?	Multiple sclerosis (MS) is a condition in which the immune system attacks the protective covering of nerve fibres (myelin) in the brain and spinal cord. This disrupts the signals travelling through the nerves. Cognitive problems occur in 50 to 70% of people with MS at some point during the disease. The most commonly affected area is information processing speed, meaning tasks and thinking that used to feel easy now take longer. Memory, attention, and word finding are also commonly affected. Many people with MS describe this as 'brain fog'.
What it affects	Processing speed (most affected), memory (verbal and visuospatial), attention and concentration, word finding. Fatigue is a major factor that worsens cognitive performance. Physical symptoms like vision problems, numbness, or weakness are separate but important context.
Screening tools	BICAMS (Brief International Cognitive Assessment for Multiple Sclerosis, Benedict et al. 2012, BMC Neurol 12:55), SDMT (Symbol Digit Modalities Test, the gold standard for MS cognitive screening), MACFIMS battery

Questions

Answer Options and Scores	Never = 0	Occasionally (1-2x/month) = 1	Frequently (weekly) = 2	Getting noticeably worse = 3

#	Question	Never (0)	Occasionally (1)	Frequently (2)	Getting worse (3)
MS1	I take noticeably longer to process what people say or to finish tasks I used to do quickly.	[ ]	[ ]	[ ]	[ ]
MS2	My thinking gets much worse when I am tired. I often call it 'brain fog'.	[ ]	[ ]	[ ]	[ ]
MS3	I forget things I was told recently, such as names, instructions, or appointments.	[ ]	[ ]	[ ]	[ ]
MS4	I lose my train of thought easily, even when it is quiet and I am not distracted.	[ ]	[ ]	[ ]	[ ]
MS5	I often struggle to find words, even for common things I know well.	[ ]	[ ]	[ ]	[ ]
MS6	I have had episodes of blurred or double vision, or loss of sight in one eye.	[ ]	[ ]	[ ]	[ ]
MS7	I have had episodes of numbness, tingling, or weakness in my arms or legs. **[RED FLAG]**	[ ]	[ ]	[ ]	[ ]
MS8	I feel very tired most of the time, even when I have not done much.	[ ]	[ ]	[ ]	[ ]
MS9	I find it hard to read, follow a film, or keep up with a conversation.	[ ]	[ ]	[ ]	[ ]
MS10	I struggle to remember things I was taught recently, such as a new process or a set of instructions.	[ ]	[ ]	[ ]	[ ]

Red Flag Note: Question MS7 (neurological symptoms like numbness, tingling, or weakness) is a physical red flag. A score of 2 or 3 on MS7, especially when combined with cognitive symptoms, should raise the result to at least Moderate Concern and prompt a GP referral for neurological assessment. These physical symptoms, alongside cognitive difficulties, are characteristic of MS and should not be dismissed.

Domain Weights

Domain	Weight	Max Raw Score	Questions
Processing Speed	30%	9	MS1, MS2, MS9
Memory	25%	9	MS3, MS10
Attention and Concentration	20%	6	MS4, MS8
Language and Word Finding	15%	3	MS5
Physical / Neurological Signs	10%	6	MS6, MS7 (1.5x weight)

Scoring Guide			
Score Range	**Label**	**Colour**	**Recommended Action**
0 to 20	Low Concern	Green	No significant signs. Recheck in 12 months.
21 to 45	Mild Concern	Yellow	Some signs present. Mention at next GP visit.
46 to 65	Moderate Concern	Orange	Multiple signs. Book a GP appointment this month.
66 to 100	High Concern	Red	Strong signs. See a doctor soon.

References

[1] Benedict RHB, et al. Brief International Cognitive Assessment for MS (BICAMS): international standards for validation. BMC Neurol. 2012;12:55. PMC: https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3607849/

[2] Langdon DW, et al. Recommendations for BICAMS. Mult Scler. 2012;18(6):891-898. PMC: https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3546642/

[3] Potticary H, Langdon D. Systematic Review and Meta-Analysis of BICAMS International Validations. J Clin Med. 2023;12(2):703. PMC: https://www.ncbi.nlm.nih.gov/pmc/articles/PMC9863826/

[4] Rao SM, et al. Cognitive dysfunction in multiple sclerosis: frequency, patterns, and prediction. Neurology. 1991;41(5):685-691. DOI: 10.1212/WNL.41.5.685

How to calculate the Final Score for any disease module

1	Add up raw scores for each domain	Sum all answer values (0 to 3) for questions in that domain
2	Apply question weights	For questions marked 1.5x weight: multiply the answer value by 1.5 first
3	Normalise each domain score	domain_norm = (domain_raw / domain_max_raw) x 100
4	Apply domain weight	domain_weighted = domain_norm x (domain_weight / 100)
5	Calculate Base Score	base_score = sum of all domain_weighted scores (0 to 100)
6	Apply Age Multiplier	Under 50: x0.7 | 50 to 64: x1.0 | 65 to 74: x1.2 | 75+: x1.4
7	Calculate Final Score	final_score = min(base_score x age_multiplier, 100)
8	Check Red Flags	If any red flag question scores 2 or 3, escalate to at minimum Moderate Concern

Scoring Guide

Score Range	Label	Colour	Recommended Action
0 to 20	Low Concern	Green	No significant signs. Recheck in 12 months.
21 to 45	Mild Concern	Yellow	Some signs present. Mention at next GP visit.
46 to 65	Moderate Concern	Orange	Multiple signs. Book a GP appointment this month.
66 to 100	High Concern	Red	Strong signs. See a doctor soon.

Quick Reference Summary

Disease	Primary Domain Affected	Key Screening Tool	Validated By	Main Red Flag
Parkinson's (PDD)	Attention, visuospatial	PANDA / MoCA-PD	Kalbe et al. 2008	Visual hallucinations
Vascular Dementia	Executive, processing speed	Hachinski Scale / MoCA	Hachinski et al. 1975	Sudden cognitive decline
Lewy Body (DLB)	Fluctuating cognition	DLB Consortium Criteria	McKeith et al. 2017	Hallucinations + dream acting
Frontotemporal (FTD)	Personality, behaviour	FBI / CBI-R	Kertesz et al. 1997	Personality change under 65
Huntington's (HD)	Processing speed, executive	UHDRS / PD-CRS	Huntington Study Group 1996	Family history + chorea
Multiple Sclerosis	Processing speed, memory	BICAMS / SDMT	Benedict et al. 2012	Neurological episodes

Important: This framework is for informational and development purposes only. It is not a clinical diagnostic tool. Questions and scoring models are based on publicly available validated instruments. The composite scoring system described here has not been independently validated in a clinical trial. Before deploying any version of this publicly, independent clinical validation and ethics review are required.

For more information on validated tools, visit: PubMed (https://pubmed.ncbi.nlm.nih.gov), MoCA (https://mocacognition.com), LBDA (https://lbda.org), HDSA (https://hdsa.org), MS Society (https://www.mssociety.org.uk).